

AWS-hosted Virtual Classroom and Learning Platform

Prepared For
Smart-Internz
Cloud Practitioner

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Project Title:

AWS-hosted Virtual Classroom and Learning Platform

Category:

Cloud Deployment | Web Application | AWS Cloud Practitioner

Skills Utilized:

1. Core Python
2. Flask Framework
3. AWS EC2, S3, RDS
4. MySQL
5. HTML, CSS, JavaScript
6. Git & GitHub

1. Project Overview:

In the modern era of digital learning, building a secure, scalable, and flexible virtual classroom is crucial. This project demonstrates the integration of Flask and multiple AWS services to develop a cloud-native educational platform.

Deployed on AWS EC2, the application leverages Amazon S3 for content storage and Amazon RDS (MySQL) for user and application data. Users can register, log in, and access course materials hosted on the cloud – making the platform both effective and expandable for future needs.

2. Key Features:

- i. Scalable Infrastructure using EC2
- ii. Secure File Storage using S3
- iii. User Management via RDS (MySQL)
- iv. Responsive Web UI for students and instructors
- v. Seamless Cloud Integration with GitHub

3. Architecture:

```
project/  
├─ app.py  
├─ templates/  
│   ├─ home.html  
│   ├─ register.html  
│   ├─ login.html  
│   └─ content.html  
└─ static/ # If needed for CSS, JS, or images
```

4. Final Project Flow:

1. Create an AWS Account:
 - a. Sign up and verify your account.

- b. Explore the AWS Management Console.

2. Create an S3 Bucket and Upload Data:

- a. Create a bucket (e.g., aws-classroom-content).
- b. Upload files (PDFs, videos).
- c. Set proper permissions (public or signed URLs).

3. Create an RDS Instance (MySQL):

- a. Launch RDS with MySQL engine.
- b. Configure DB instance and create a database.
- c. Connect using MySQL Workbench to create tables.

4. Launch and Configure EC2 Instance:

- a. Launch instance with Amazon Linux 2 or Ubuntu.
- b. Set security groups and SSH key pair.
- c. Install Python, Flask, MySQL client.

5. Develop Flask App:

- a. Build routes for register, login, content.
- b. Create templates: home.html, register.html, login.html, content.html.
- c. Use Bootstrap for styling.
- d. Connect app to AWS S3 (using boto3) and RDS.

6. Deploy Flask App on EC2:

- a. SSH into EC2.
- b. Clone GitHub repository. • Install dependencies: `pip install -r requirements.txt`.
- c. Run app using Gunicorn + Nginx (optional).

7. Upload Code to GitHub:

- a. Create repository.
- b. Push project files with commits and documentation.

8. Test Scenarios:

- a. **Scenario 1: Student Registration and Login**
- b. **Scenario 2: Admin Upload of Course Materials**

c. **Scenario 3: Downloading Course Content**

5. User Scenarios:

Scenario 1: Student Registration and Course Access

- **User:** Alice Johnson
- **Process:** Registers via form, logs in, and accesses course content from S3

Scenario 2: Admin Uploads Course Materials

- **User:** System Admin
- **Process:** Uploads PDFs; content is stored in S3 and metadata is updated in RDS.

Scenario 3: Student Downloading Course Content

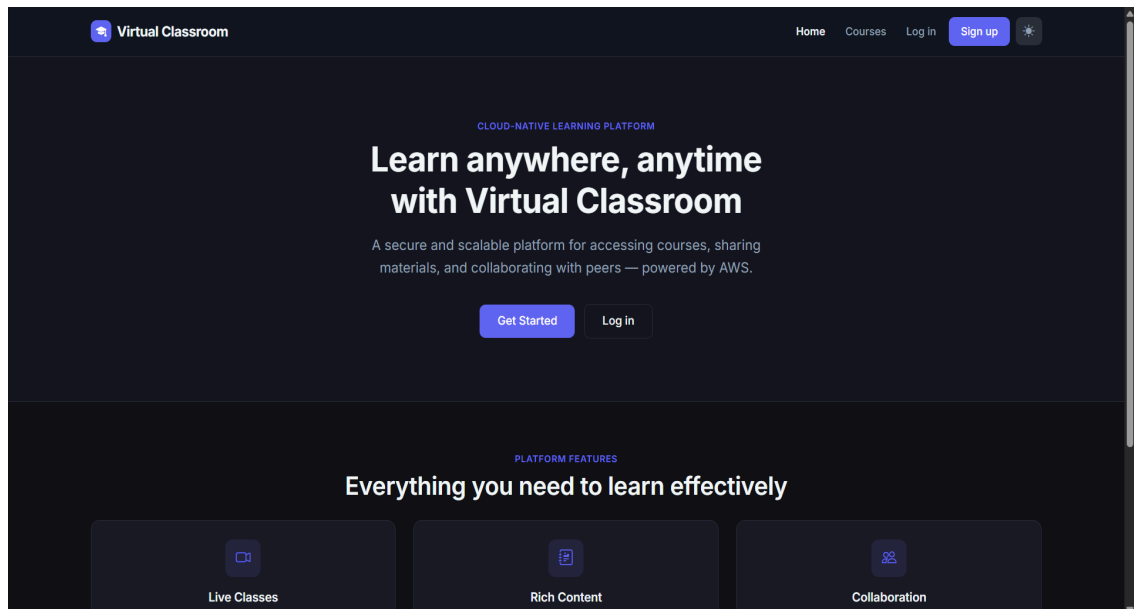
- **User:** Bob Patel
- **Process:** Selects a file, clicks a link, and downloads directly from S3.

6. Challenges Faced:

- a. Learning AWS services and IAM policies
- b. Managing AWS credentials securely
- c. Flask and AWS integration (using boto3)
- d. RDS connection issues and MySQL Workbench setup
- e. Debugging EC2 deployment issues

7. Output Pages:

- Landing page with navigation



- Registration form

The screenshot displays the registration form titled 'Create your account'. The form is set against a dark background with light-colored input fields. It includes fields for 'First name' (filled with 'Omkar'), 'Last name' (filled with 'Patil'), 'Email' (filled with 'omkarpatil03092003@gmail.com'), 'Password', and 'Confirm password'. A checkbox labeled 'I agree to the Terms and Conditions' is checked. A 'Create account' button is positioned below the form. At the bottom, a link for 'Already have an account? Log in' is visible. The header of the page matches the landing page, with the 'Sign up' button highlighted.

- Login page

The screenshot shows the login page of a web application named "Virtual Classroom". The page has a dark theme. At the top, there is a navigation bar with links for "Home", "Courses", "Log in", and "Sign up". The main heading is "Log in to your account" with a subtext "Enter your credentials to access the platform." Below this, there is a form with two input fields: "Email" (containing "omkarpatil03092003@gmail.com") and "Password" (masked with dots). There is a "Remember me" checkbox and a "Log in" button. At the bottom of the form, there is a link "Don't have an account? Sign up". The footer contains the copyright notice "© 2026 Virtual Classroom. All rights reserved."

- Course materials page with download links from S3

The screenshot shows the course materials page of the "Virtual Classroom" web application. The page has a dark theme. At the top, there is a navigation bar with links for "Home", "Courses", "omkarpatil03092003", and "Log out". The main heading is "Available Courses" under the "BROWSE" section. Below this, there are three course cards: "Web Development" (Beginner level), "Data Science" (Intermediate level), and "Mobile Development" (Advanced level). Each card has an "Enroll" button. Below the course cards, there is a section titled "Course Materials" under the "RESOURCES" section.

8. Conclusion:

This project highlights the integration of cloud computing with web development to build a fully operational virtual classroom. Leveraging AWS's scalability and Flask's simplicity, the platform achieves reliable user access, secure data handling, and an overall smooth educational experience.

9. GitHub And Demo Link:

Demo Link :

<https://youtu.be/Txw708C59RQ>

GitHub Link:

<https://github.com/omkarpatil3903-hub/aws-hosted-virtual-classroom->